essay v1 final review 3

This is a well-constructed piece — the history checks out. I searched to confirm the load-bearing facts, and they're all real: Chui's 1971 Monthly note, Newman's 1972 $\pi/18$ bound, the 2021 Abakumov–Borichev–Fedorovskiy paper solving the Hilbert-space version in *Math. Ann.* — all correct. And I was genuinely surprised the Miftakhov item checked out too: Meduza, The Insider, and the Moscow Times all confirm that Bernard Randé sent Azat Miftakhov a Hilbert-space variant of Chui's conjecture, that he solved it in prison over about four months, and that it appeared in *Revue de la filière mathématique* in August 2026. Your essay's own hedge ("not independently verified," "should still be regarded as open for $N \geq 3$") is exactly right, since the reporting is about a Hilbert-space analog, not the original L^1 statement — good instinct to flag that distinction rather than let the headline blur it.

My own numerical check. I reran an independent quadrature + multi-start Nelder–Mead (Gauss–Legendre in r, midpoint in θ, 20 restarts) rather than trust your numbers blind:

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N=2: equispaced=1.62798, optimizer best=1.62798  (exact match)
N=3: equispaced=1.80247, optimizer best=1.80247  (exact match)
N=4: equispaced=1.90557, optimizer best=1.90611  (0.03% — quadrature/optimizer
    ↪noise)
N=5: equispaced=1.97329, optimizer best=1.97490  (0.08% — same)
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This corroborates your §7: no configuration search beats equispaced, and the tiny "excess" values track quadrature resolution rather than any real competitor. It's the same story, independently obtained.

My idea, for what it's worth. The essay is honest that the ABF proof lives entirely on the pairwise-kernel expansion available in Hilbert space, and that this dies in L^1. Here's a route I'd try that doesn't need a pairwise expansion: a **layer-cake + circular-symmetrization argument**, in the spirit of Baernstein's star-function technique.

Write $\|f\|_{L^1} = \int_0^\infty m_2(\{z \in \mathbb{D} : |f(z)| > t\})\, dt$ (layer cake). Since f is holomorphic, $|f|$ is subharmonic away from the poles, so the super-level sets $\{|f| > t\}$ obey a maximum-principle-type rigidity that Baernstein's circular symmetrization is built to exploit: for subharmonic u on the disc, the *symmetric decreasing rearrangement of* u *along circles* — its "star function" $u^*(re^{i\theta}) = \sup\{$ measure of the set on $|z| = r$ where $u \geq$ the value attained by u^*'s own construction $\}$ — dominates u pointwise in a way that controls $\int u\, dm_2$. The natural conjecture is that *polarizing* two poles toward antipodal points on $\mathbb{T}$ (the two-point circle symmetrization used throughout that theory) can only *decrease* $m_2(\{|f| > t\})$ for every t simultaneously — which would give equispacing as the minimizer by a standard iterative-polarization compactness argument, without ever expanding $|f|$ into pairwise terms.

The reason I think this has a chance where naive attempts fail: your §6 sharpness result (Theorem 1.3, oppositely-signed charges can cancel to near-zero) is really a statement that *cancellation* is a same-vs-opposite-sign phenomenon, and polarization is exactly the operation that's known to respect sign-coherent superposition (it's literally built from reflecting mass across a diameter, which is the same-sign case). The place I'd expect it to break is that $|f|$'s level sets aren't simply connected once $N \geq 3$ — the "many small islands" topology of $\{|f| > t\}$ near each pole is exactly what defeats single-variable differentiation-under-the-integral in Nazarov's proof, and polarization comparisons usually need some topological control on the level sets to run cleanly. So this isn't a proof sketch so much as: *if polarization monotonicity for super-level sets of* $|f|$ *can be established even in a weak (single-swap, not full symmetrization) form, N ≥ 3 falls out by compactness* — and that monotonicity claim is a concrete, checkable sub-question, possibly even numerically first (compare $m_2(\{|f| > t\})$ before/after swapping one pole pair toward antipodal, across many random configurations and many t).

That's speculative, but it's a genuinely different mechanism from the three eras you identify (geometric/local, one-parameter/analytic, structural/convex) — it would be a *rearrangement* era — and it's the kind of thing worth a numerical pilot before investing in the analysis.